



Algorithm & C-Programming

Chapter 00: Syllabus

For the engineering school: ECE-Paris



Metadata

Edition:

1.0

 (2025.10.09)

Designed for the open standard:

C23

 (2024)

Written by: Simon Céleste VANDAMME

svandamme@omnesintervenant.com



01. FOREWORD

A. General Information

This document should take about: **10mn**
to be read.

B. Prerequisite Chapters

These are the recommended chapters to read:

- **NONE!**
This chapter doesn't require prior knowledge in C.



C. AI: Recommendation and Downsides

Artificial Intelligence (AI) tools show remarkable results, especially in the field of programming. Their integration into most Integrated Development Environments (IDEs) nowadays makes them more accessible than ever.

Used wisely, AI can be a valuable support in your journey to learn C programming. However, it can also be tempting to rely on it to complete entire projects or exercises in seconds.

When learning a new skill in any subject, remember this golden rule:

UNDERSTANDING > REWRITING PRE-DONE CONTENT

Note that I am NOT AGAINST the use of AI in your works.

However, I would highly appreciate it if you were transparent upon using it in your code and specify EVERY TIME why you are relying on it and how the snippet code works.

RECOMMENDATIONS

01. First think about the subject YOURSELF, write some elements down.
Then, use AI to continue your work if needed.

Your learning suffers when you let AI solve problems before trying yourself.
Skip the thinking, and you only train dependency... not skill.

As future engineers, you are expected to understand problems deeply and think creatively.
Keep training those skills: they are the foundation of your value as an engineer.

02. Use AI tools as personal teachers.
They are great to highlight your mistakes or improve your code quality.

Once done with a working program, you can prompt the AI on what is to be improved, for example:

- How to secure your code against errors?
- Ask for alternative ways to solve the problem. Did you use the most interesting way?
- How to improve your memory management or time execution?

03. YOU are responsible for the code you submit... not the AI.

Whether at school or in your future job, your program is your creation. You must fully understand it and be able to explain every part of it, even if you received significant help from AI.

AI has limits: it struggles with complex projects, can produce insecure code, and lacks real creativity.
As an engineer, it is your job to go beyond these limits.

02. TABLE OF CONTENTS

01. FOREWORD	2
A. General Information	2
B. Prerequisite Chapters.....	2
C. AI: Recommendation and Downsides	3
02. TABLE OF CONTENTS	4
03.SYLLABUS.....	5
A. General Overview	5
B. Detailed Syllabus	6
01. The Programmer’s Essentials	6
02. Advanced Programmer Courses	9
03. TECHNICAL SPECIFICATION SHEET	11
A. REFERENCES	11
B. DOCUMENT SPECIFICATIONS	12



03. SYLLABUS

A. General Overview

The Programmer's Essentials		
ID	Module	Title
TPE-001	M1 Foundations of Programming	Introduction & A Bit of History About the C Language
TPE-002		Understanding How Programs Work
TPE-003		Getting Started with C
TPE-004	M2 Core Syntax & Data Handling	The Basics of C
TPE-005		Control Flow
TPE-006		Functions
TPE-007		Random Number Generation
TPE-008	M3 Memory & Structures	The Pointers: The Core of C
TPE-009		Arrays: From 1D to 3D... and Beyond!
TPE-010		Dynamic Memory Management
TPE-011	M4 Debugging & Modular Design	Debugging and Troubleshooting
TPE-012		Modular Programming & Libraries
TPE-013	M5 Advanced C	Recursion
TPE-014		Structures and Advanced Types
TPE-015		File Handling
TPE-016	M6 Project Work	Best Practices and Code Style in C
TPE-017		Capstone Project: Managing a Software Project

Advanced Programmer Courses		
ID	Module	Title
APC-001	M1 System Programming & Concurrency	Networking with Sockets
APC-002		Multithreading & Concurrency in C
APC-003		Operating Systems Concepts in C
APC-004	M2 Real Time Programming & Engine Design	SDL3: Creating Something VISUAL (Graphical Library)
APC-005		Building a Game Loop and Simulation Engine
APC-006	M3 Data & Algorithm	Data Structures & Algorithms in Depth
APC-007		Data Persistence & Serialization
APC-008	M4 Code Quality & Tooling	Unit Testing & Continuous Integration in C
APC-009		Advanced Debugging & Profiling
APC-010		Building Cross-Platform C Libraries
APC-011	M5 Language Design & Interpretation	Interfacing C with Other Languages (Python, C++, etc.)
APC-012		Compilers & Interpreters: Building a Mini Language



B. Detailed Syllabus

This is your advanced roadmap on the journey to mastering the C-language.

Chapters are grouped into *Modules* to help you easily find your way, whether you're revisiting past lessons or looking ahead to what's next.

01. The Programmer's Essentials

Module 1 - Foundations of Programming

TPE-001 Introduction & A Bit of History About the C Language

Discover where the C language comes from, why it was created, and how it shaped modern programming.

Learn about its legacy in UNIX, operating systems, and beyond.

TPE-002 Getting Started with C

Explore how computers execute code, from source files to binary instructions. Understand compilation, linking, and the relationship between software and hardware.

TPE-003 Understanding How Programs Work

Set up your development environment, learn how to compile and run your first C program, and get familiar with syntax, comments, and structure.

Module 2 - Core Syntax & Data Handling

TPE-004 The Basics of C

Learn the fundamental building blocks of programming: data types, variables, constants, operators, and basic input/output using *printf* and *scanf*.

TPE-005 Control Flow

Master conditional statements (if, switch) and loops (for, while, do...while) to control how your programs make decisions and repeat tasks.

TPE-006 Functions

Understand how to structure your code using functions, pass arguments, return values, and manage scope and lifetime of variables.

TPE-007 Random Number Generation

Learn to create dynamic, unpredictable behavior using C's pseudo-random functions. Discover *rand*, *srand*, and time-based seeding.



Module 3 - Memory & Structures

TPE-008 The Pointers: The Core of C

Learn how pointers work, how to manipulate memory addresses, and why they're essential for dynamic memory, arrays, and efficient data handling.

TPE-009 Arrays: From 1D to 3D... and Beyond!

Dive into one-dimensional and multi-dimensional arrays.
Explore array manipulation, iteration, and how arrays interact with pointers.

TPE-010 Dynamic Memory Management

Learn how to allocate, reallocate, and free memory using *malloc*, *calloc*, and *free*.
Understand memory leaks and how to prevent them.

Module 4 - Debugging & Modular Design

TPE-011 Debugging and Troubleshooting

Develop essential debugging skills using practical tools and techniques to find logic errors, segmentation faults, and runtime issues efficiently.

TPE-012 Modular Programming & Libraries

Organize your projects into reusable modules.
Learn to separate code into *header* (.h) and *source* (.c) files, and to link static or dynamic libraries.



Module 5 - Advanced C

TPE-013 Recursion

Understand recursive thinking and apply it to solve mathematical and algorithmic problems, like factorials, Fibonacci, and tree traversals.

TPE-014 Structures and Advanced Types

Learn to define and use *struct*, *typedef*, and nested data structures to model complex information and create more powerful programs.

TPE-015 File Handling

Learn how to read, write, and manage files in C using file pointers.
Handle structured data, configuration files, and persistent storage.

Module 6 – Project Work

TPE-016 Best Practices and Code Style in C

Discover how to write clean, maintainable, and portable C code.
Focus on naming conventions, documentation, formatting, and project readability.

TPE-017 Capstone Project: Managing a Software Project

Apply all your skills in a full-scale project. Plan, design, code, and debug a complete C application, while learning teamwork and version control basics.

02. Advanced Programmer Courses

Module 1 – System Programming & Concurrency

APC-001 Networking with Sockets

Learn the basics of network communication using sockets.
Build client-server applications and understand protocols like TCP/IP.

APC-002 Multithreading & Concurrency in C

Master the fundamentals of parallel programming using POSIX threads.
Handle synchronization, shared data, and performance optimization.

APC-003 Operating Systems Concepts in C

Build a deeper understanding of OS principles: process management, memory allocation, file systems, and scheduling... all through C examples.

Module 2 – Real Time Programming & Engine Design

APC-004 SDL3: Creating Something VISUAL (Graphical Library)

Learn to create interactive, graphical programs using SDL3.
Explore event-driven programming, rendering, and real-time input handling.

APC-005 Building a Game Loop and Simulation Engine

Understand how real-time simulations and games are structured.
Learn about timing, event queues, frame updates, and modular design... the logic backbone of any interactive program.

Module 3 – Data & Algorithm

APC-006 Data Structures & Algorithms in Depth

Implement and analyze classic structures (linked lists, trees, graphs, hash tables) and algorithms (sorting, searching, pathfinding) in C.

APC-007 Data Persistence & Serialization

Learn how to store and retrieve structured data efficiently.
Explore binary file formats, text encodings, and serialization techniques to save complex data structures such as game states, configurations, and logs.

Module 4 – Code Quality & Tooling



APC-008 Unit Testing & Continuous Integration in C

Discover how to write automated tests, use testing frameworks, and integrate CI pipelines for reliable, professional-grade software.

APC-009 Advanced Debugging & Profiling

Learn to use tools like GDB, Valgrind, and profilers to inspect performance, memory usage, and execution flow in large codebases.

APC-010 Building Cross-Platform C Libraries

Learn to design, document, and compile reusable C libraries for multiple operating systems, including static and dynamic linking techniques.

Module 5 – Language Design & Interpretation

APC-011 Interfacing C with Other Languages (Python, C++, etc.)

Learn how C libraries can communicate with higher-level languages.
Explore using C with Python (via CFFI), C++, or shared libraries to build powerful, hybrid applications.

APC-012 Compilers & Interpreters: Building a Mini Language

Step into the world of language design.
Write a simple parser and interpreter in C, and understand how programming languages work internally.



03. TECHNICAL SPECIFICATION SHEET

A. REFERENCES

The following references and resources were used as guidance and inspiration in the development of the courses.

Books:

- “Apprenez à programmer en C”, 2012, Mathieu NEBRA
- “Algorithms for Dummies”, 2024, John Paul Mueller and Luca MASSARON

Websites / Online documentation:

- WordReference: <https://www.wordreference.com/> *Translation: French <> English*

Software / Tools:

- **IDE (C programming):** *Used for screenshots, programming and testing.*
 - CLion (*JetBrains*)
 - Code::Blocks
 - Visual Studio Code (*Microsoft*)
- **Artificial Intelligence:** *Help in designing the syllabus and create original exercises.*
 - ChatGPT (*OpenAI*)
 - Gemini (*Google*)
- **Writing/Editing software:** *Creating my courses, exercises, presentations...*
 - PowerPoint (*Microsoft*)
 - Word (*Microsoft*)



B. DOCUMENT SPECIFICATIONS

TECHNICAL INFORMATION	
Document Version	1.0
Release Date (YYYY.MM.DD)	2025.10.09
Course Title	ALGORITHM & C-PROGRAMMING
Chapter Name	Chapter 0 - Syllabus
Document ID	C-SYLLABUS-001
Keywords	C, IDE, program, worksheet, course materials, digital
File Name	C-Syllabus-v1.0.pdf
Page Count	12 pages
Contributors	▪ Nathalie AMSALLEM: Proofreader ▪ Léo BELUITE: Proofreader ▪ Drazic MARTIN: Proofreader and program tester ▪ Simon Céleste VANDAMME: Author
Maintainer Contact	celeste.vandamme@outlook.com
Target Audience	Designed for engineering students starting to learn the C-programming language.
Languages Available	▪ English: available ▪ French: not yet available
Answer Sheet	Not required (no exercises)
Confidentiality Level	Public
License	Creative Commons Attribution 4.0 International (CC BY 4.0) Link: https://creativecommons.org/licenses/by/4.0/



VERSION HISTORY			
Version	Release Date (YYYY.MM.DD)	Authors	Description / What is new in the version?
1.0	2025.10.09	- VANDAMME S.	Official release of the document.
1.1	2025.10.09	- VANDAMME S.	Fixed blue highlight on module + fix on a wrong course ID number